

Parixit Sanghani

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Profile

Software Engineer with experience designing and automating large-scale distributed systems and data pipelines in cloud environments. Skilled in multiple programming languages, with expertise in backend distributed architecture, microservices, cloud automation, and infrastructure-as-code. Passionate about building reliable, high-performance systems for data-driven products.

Work Experience

Amazon Robotics

07/2024 - 10/2025

Software Developer Engineer

Boston, USA

- Engineered a **cloud-native MLOps pipeline** utilizing **Docker** for a high-performance **PyTorch** model, resulting in a **99.9% uptime** for real-time inference and providing scalable resource allocation.
- Developed backend infrastructure using **AWS, Step Functions, DynamoDB, EMR, Docker, and EMR**, reducing model training time by **30%** and cloud infrastructure costs by **\$1.1M** annually.
- Designed and implemented a robust artifact versioning system leveraging **Distributed ECS, S3, and DynamoDB** to manage and deploy machine learning models, accelerating the release cycle for research teams by **25%**.
- Improved development productivity by building internal CI/CD tooling using **AWS CDK, CloudFormation, and CodeBuild**, reducing build & deployment time by **50%**.
- Built cost-efficient distributed pipelines for ML-powered robotics using serverless architecture (e.g., **Java Lambda functions, SQS**), reducing operating costs by **\$50K** per month while ensuring high reliability for global deployments.

Amazon Robotics

01/2023 - 08/2023

Software Developer Engineer Co-op

Boston, USA

- Created scalable microservices for robotics simulation using **Spring Boot (Gradle), Kotlin** and **AWS**, improving modularity and maintainability.
- Enhanced system reliability and accelerated development cycles by **20%** through integration, unit testing and A/B testing.
- Designed and executed simulation pipelines via custom APIs and automation scripts, streamlining experimentation workflows for applied research teams.
- Implemented serverless functions (e.g., **AWS Lambda, Python**) and comprehensive unit tests integrated with other applications to process Data Lake ingestion events, reducing data processing latency by **45%** for customers.

Crest Data

12/2019 - 10/2021

Software Developer Engineer

Ahmedabad, India

- Architected full-stack log ingestion ETL pipelines using **Python** and **ReactJS** to automate the processing and visualization of large-scale Google Chronicle security datasets for enterprise clients, improving data visibility and analysis speed by **60%**.
- Developed and deployed a ReactJS interface for Cisco's competitive analysis tool, enabling **2,000+** vendors to identify relevant products **40%** faster.
- Built GraphQL endpoints for the internal HRMS, cutting query payload size by **50%** and improving network performance.
- Modernized legacy HashiCorp-based and custom React components by refactoring UI logic, improving performance, and upgrading outdated frontend modules, resulting in faster load times and more maintainable code.

Technical Skills

Programming Languages: Python, Java, Kotlin, TypeScript, Go, Rust, C++, Shell Scripting, PL/SQL, Linux

Frameworks & Libraries: Spring Boot, Django, Node.js, React.js, Flask, FastAPI, NGINX, Kafka, Redis, gRPC, REST

Databases: MySQL, PostgreSQL, DynamoDB, MongoDB, Oracle Database, Microsoft SQL Server

Cloud & DevOps: AWS, GCP, Azure, Docker, Kubernetes, Terraform, CloudFormation, Jenkins, CI/CD Pipelines

Software Development Practices: OOP, Distributed Systems, Pipeline Automation, Microservices, Code review, SDLC, Debugging

Certifications: AWS training and certification, PathAI Deep Learning, Machine Learning - Coursera, Python - IITM

Education

Rochester Institute of Technology

Rochester, NY

Master of Computer Science

01/2022 - 08/2024

Technical Projects

- Intelligent Document Extraction (2025):** Designed an end-to-end **OCR + GenAI** system to parse unstructured **PDFs**, transform extracted insights into normalized SQL records, and deliver real-time data to the frontend through robust API integrations.
- Telemetry & Data Pipeline Automation (2024):** Implemented Python microservices backed by **Kafka** to reliably stream and visualize **Kubernetes** pod metrics, significantly improving observability and end-to-end ML pipeline traceability.
- Efficient HPC Storage (2024):** Re-architected the Lustre filesystem into Kubernetes-based microservices with **self-healing** and **pod recovery**, improving scalability, availability, and deployment efficiency in cloud environments.